

Random Sampling and Simple Distribution

Aim:

To perform random sampling and demonstrate the concept of sampling distribution using Python.

Procedure:

1. Import the required libraries — NumPy for calculations and Matplotlib for visualization.
2. Generate a large population dataset (e.g., 10,000 values) following a normal distribution.
3. Define the population's mean and standard deviation parameters.
4. Select a random sample of a fixed size (e.g., 50) from the population without replacement.
5. Plot histograms to compare the population distribution and the sample distribution.
6. Define the number of samples to draw (e.g., 1000) for building a sampling distribution.
7. For each iteration, randomly select a sample and compute its mean.
8. Store all sample means to form the sampling distribution of the mean.
9. Plot the histogram of the sample means to visualize the sampling distribution.
10. Calculate and print the population mean, mean of sample means, and their standard deviations to observe the Central Limit Theorem effect.

```
import numpy as np
import matplotlib.pyplot as plt

population = np.random.normal(loc=70, scale=10, size=10000)

sample_size = 50
sample = np.random.choice(population, sample_size, replace=False)

plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.hist(population, bins=30, edgecolor='black')
plt.title("Population Distribution (Mean = {:.2f})".format(np.mean(population)))
plt.xlabel("Values")
plt.ylabel("Frequency")

plt.subplot(1, 2, 2)
plt.hist(sample, bins=15, edgecolor='black', color='orange')
plt.title("Single Random Sample (n = {})".format(sample_size))
plt.xlabel("Values")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()

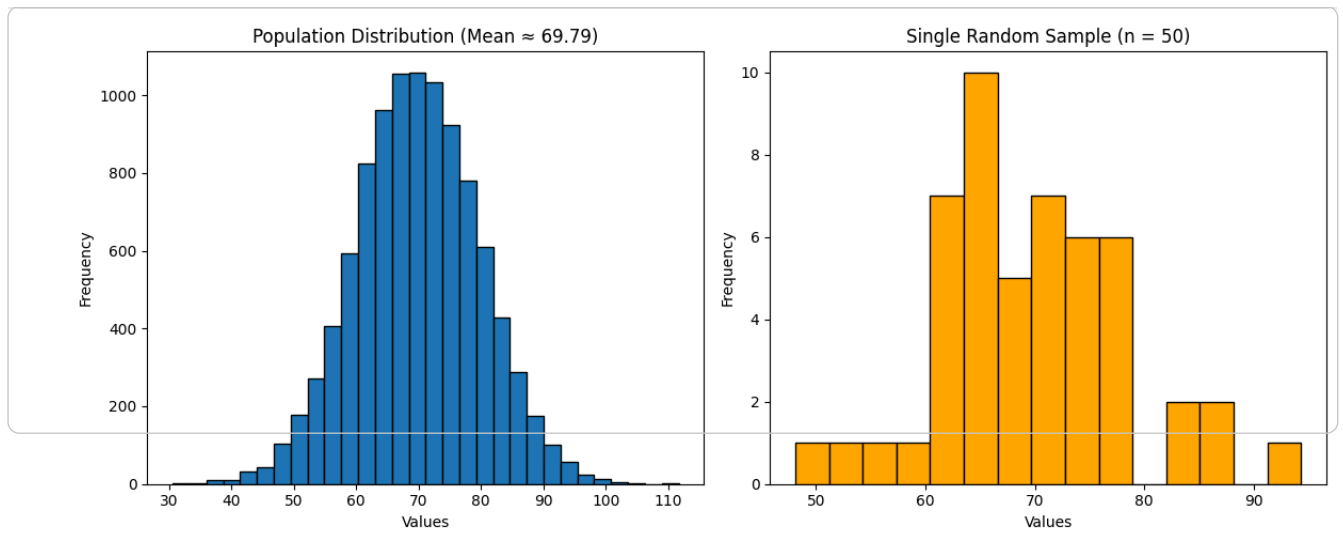
num_samples = 1000
sample_means = []

for i in range(num_samples):
    s = np.random.choice(population, sample_size, replace=False)
    sample_means.append(np.mean(s))

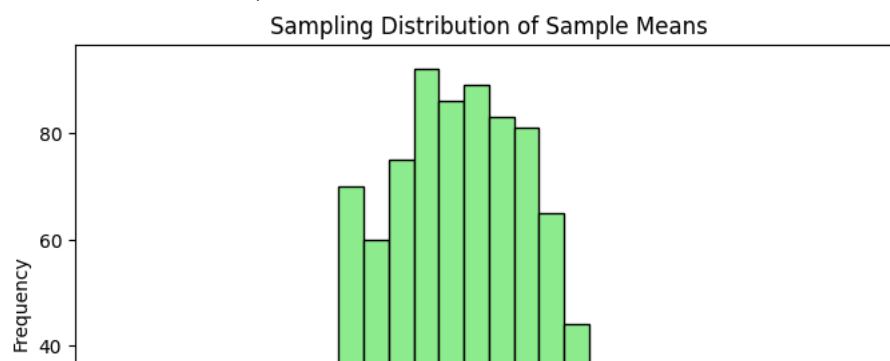
plt.figure(figsize=(8,5))
plt.hist(sample_means, bins=30, color='lightgreen', edgecolor='black')
plt.title("Sampling Distribution of Sample Means")
plt.xlabel("Sample Mean Values")
plt.ylabel("Frequency")

print("Population Mean:", np.mean(population))
print("Mean of Sample Means:", np.mean(sample_means))
print("Standard Deviation of Population:", np.std(population))
print("Standard Deviation of Sample Means:", np.std(sample_means))

plt.show()
```



Population Mean: 69.79483907071406
Mean of Sample Means: 69.76640121126027
Standard Deviation of Population: 9.90156353815207
Standard Deviation of Sample Means: 1.3756700603765846



Result

Hence, a Python program was written to perform random sampling and demonstrate that the distribution of sample means approaches a normal distribution with a mean approximately equal to the population mean, verifying the **Central Limit Theorem**.